

Supplementary Material

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Figures

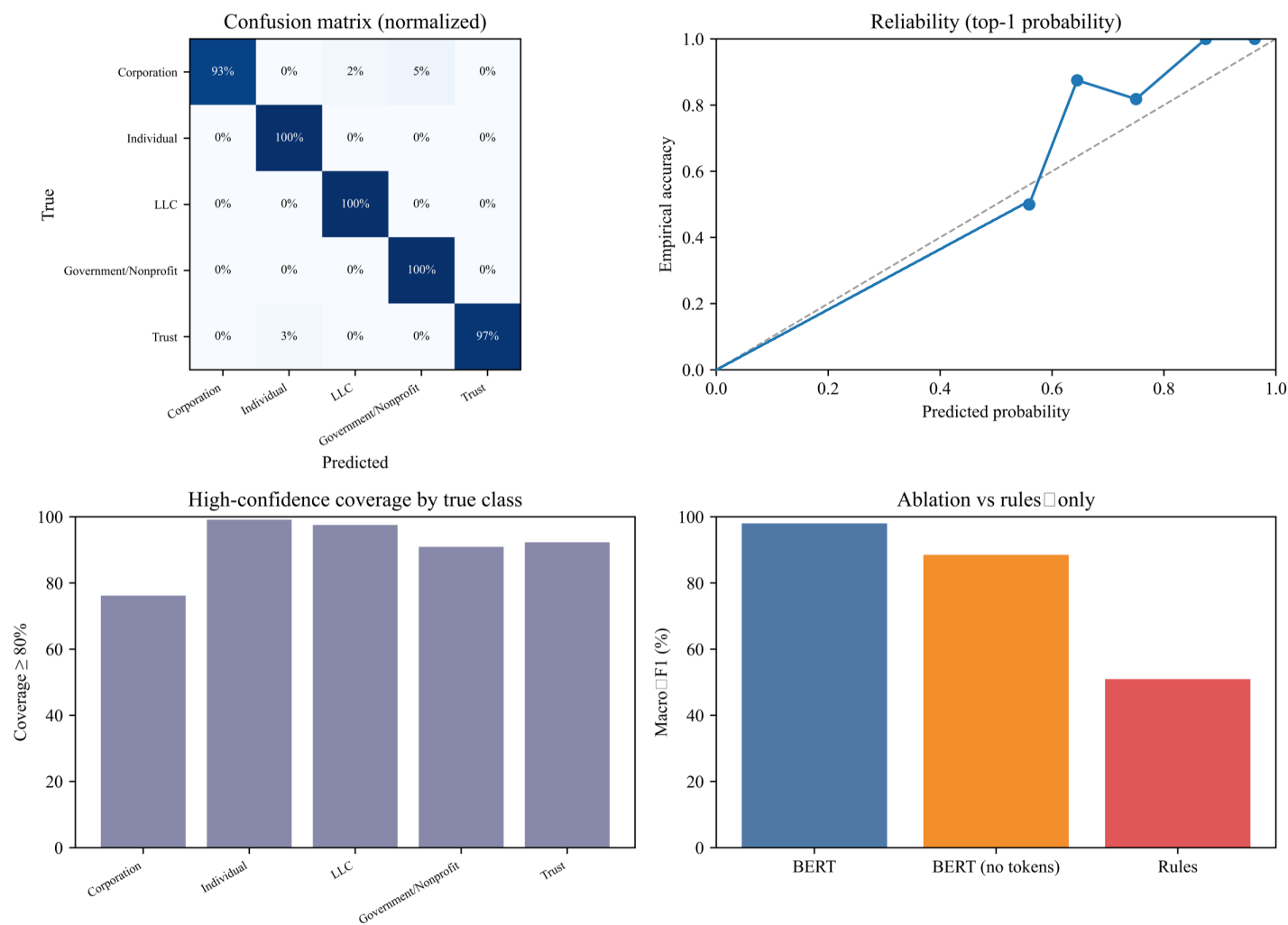
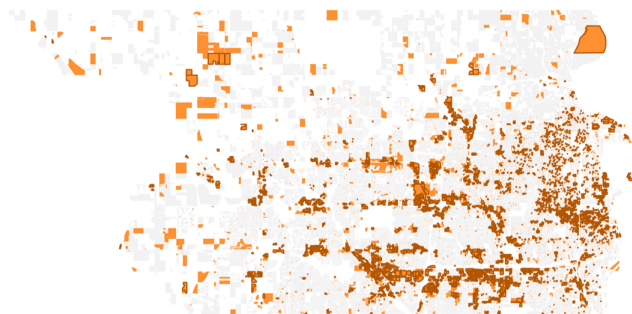


Figure 1: **Figure S1.** Classifier validation panels for the BERT model showing (a) normalized confusion matrix, (b) reliability calibration curve, (c) high-confidence coverage by class, and (d) ablation comparison.

SFHA parcels — Moran's $I=0.783$



LLC holdings — Moran's $I=0.268$



Corporation holdings — Moran's $I=0.196$



Appraisal neighborhoods (FE strata)

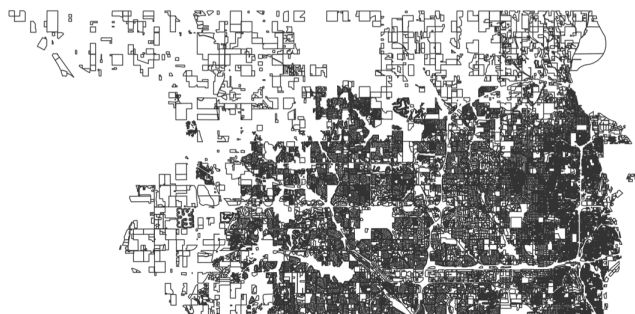


Figure 2: **Figure S2.** Map of Douglas County Single Family Residential Parcels by Entity Type

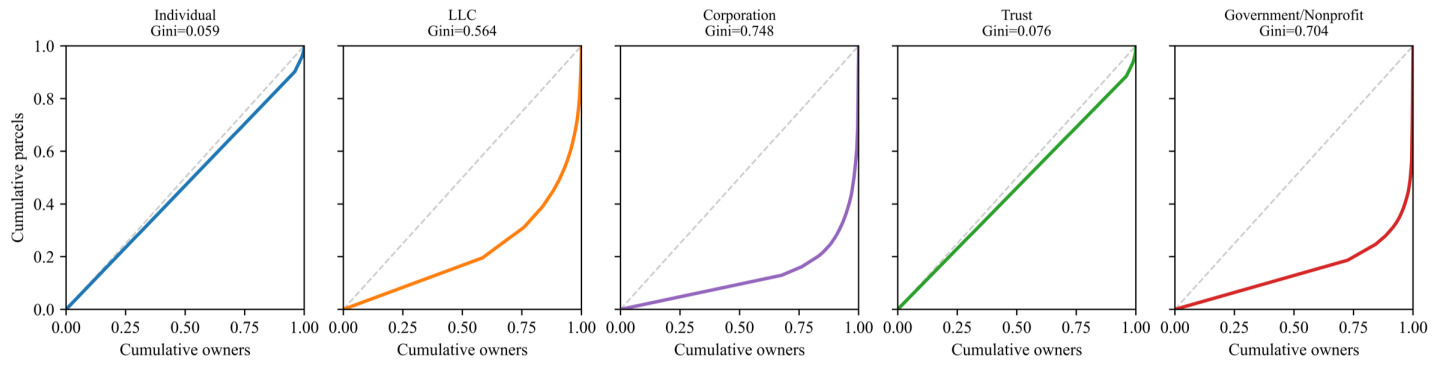
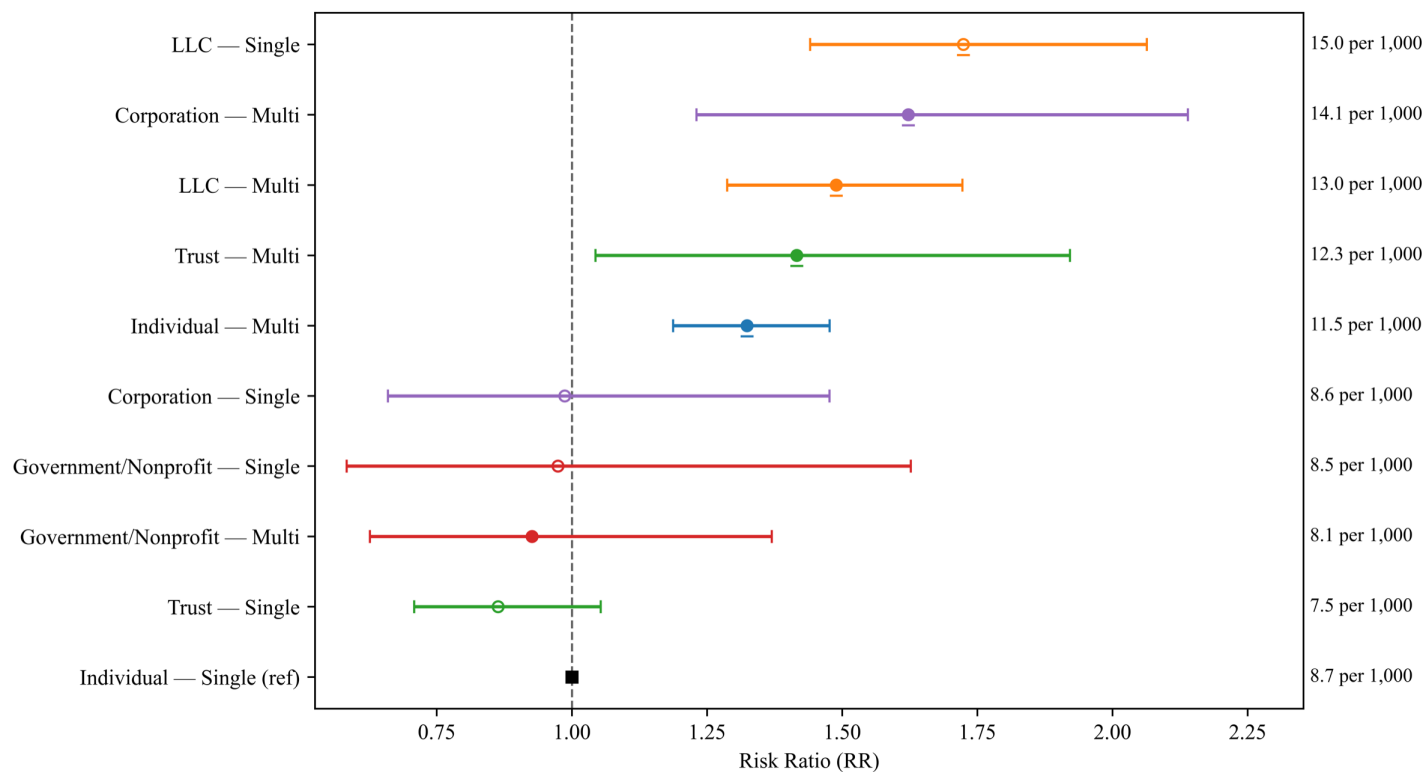


Figure 3: **Figure S3.** Concentration of single family SFHA-located parcels by organizational form



Entity LISA — Individual (SFR)

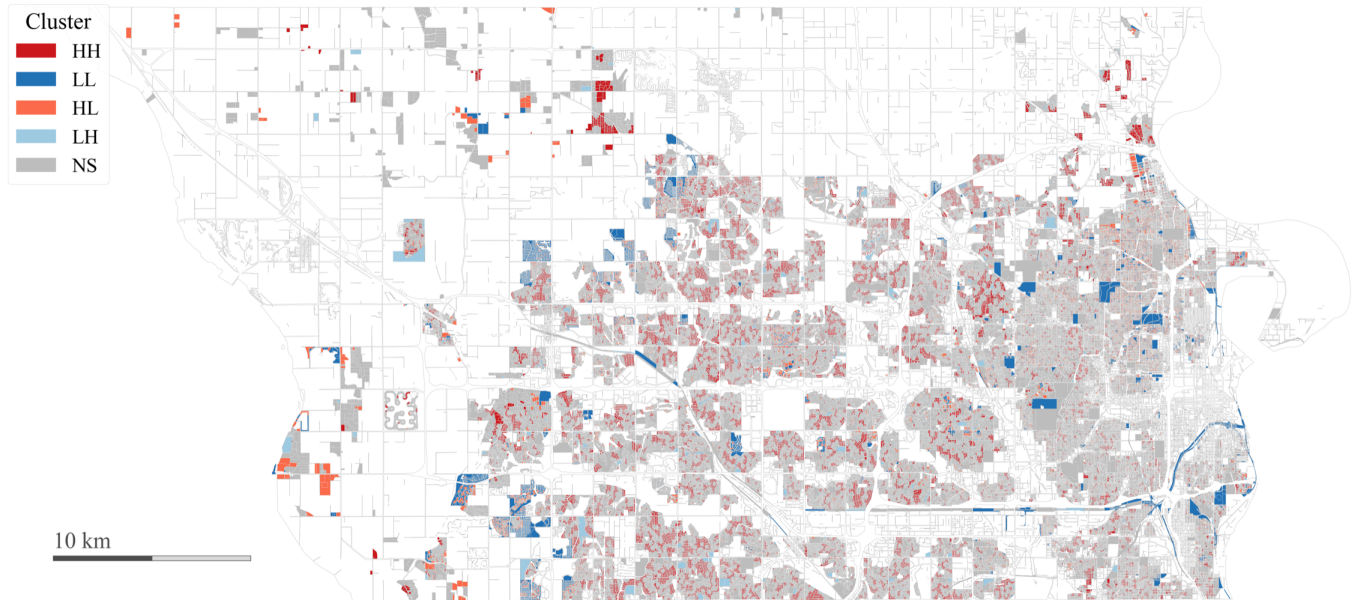


Figure 5: **Figure S5.** Individual Owner Spatial Clustering by Parcel

Entity LISA — LLC (SFR)

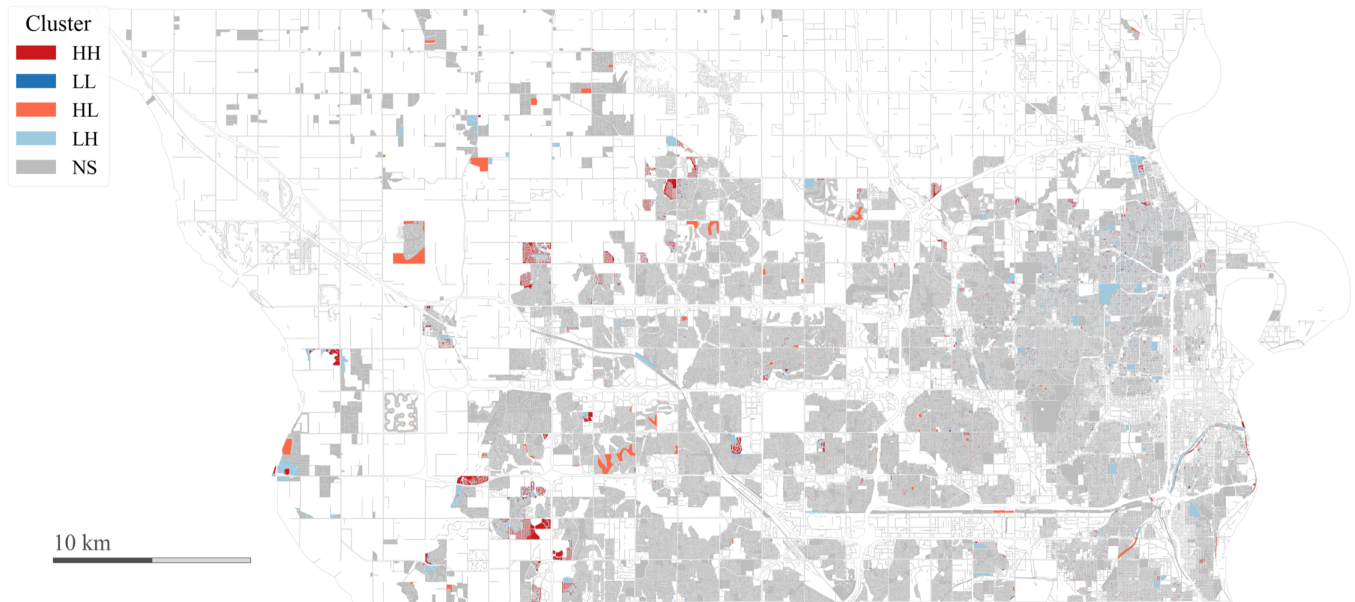


Figure 6: **Figure S6.** LLC Owners Spatial Clustering (LISA) by Parcel

Entity LISA — Corporation (SFR)

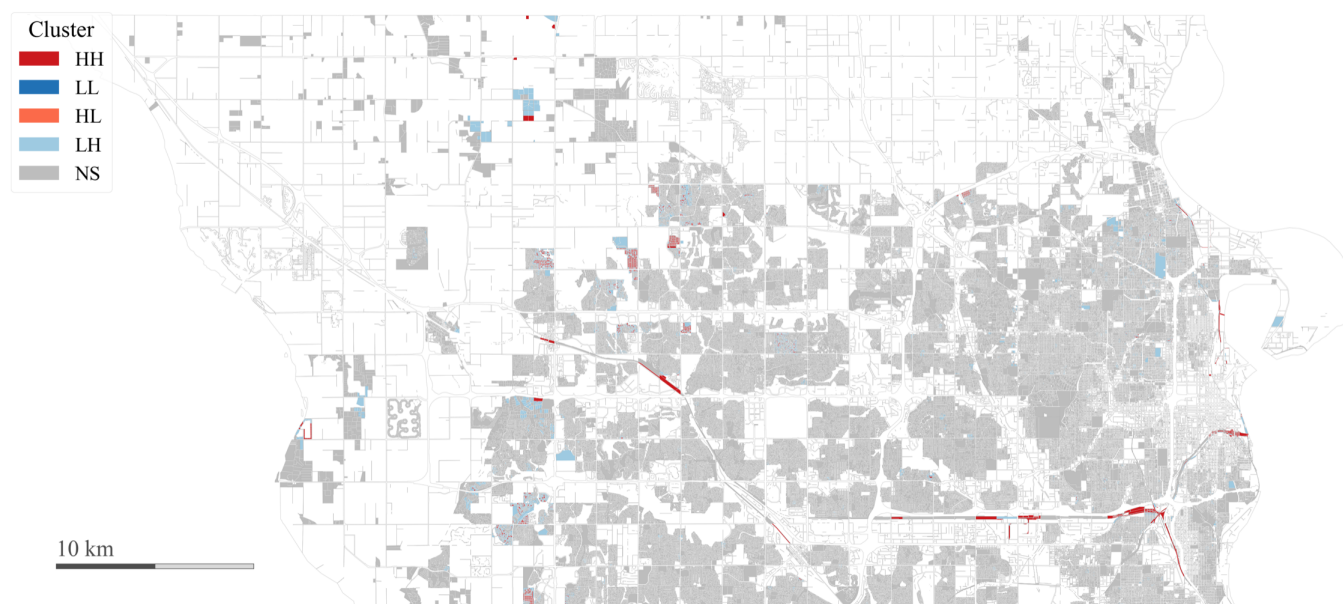


Figure 7: **Figure S7.** Corporate Owners Spatial Clustering (LISA) by Parcel

Entity LISA — Government or Nonprofit (SFR)

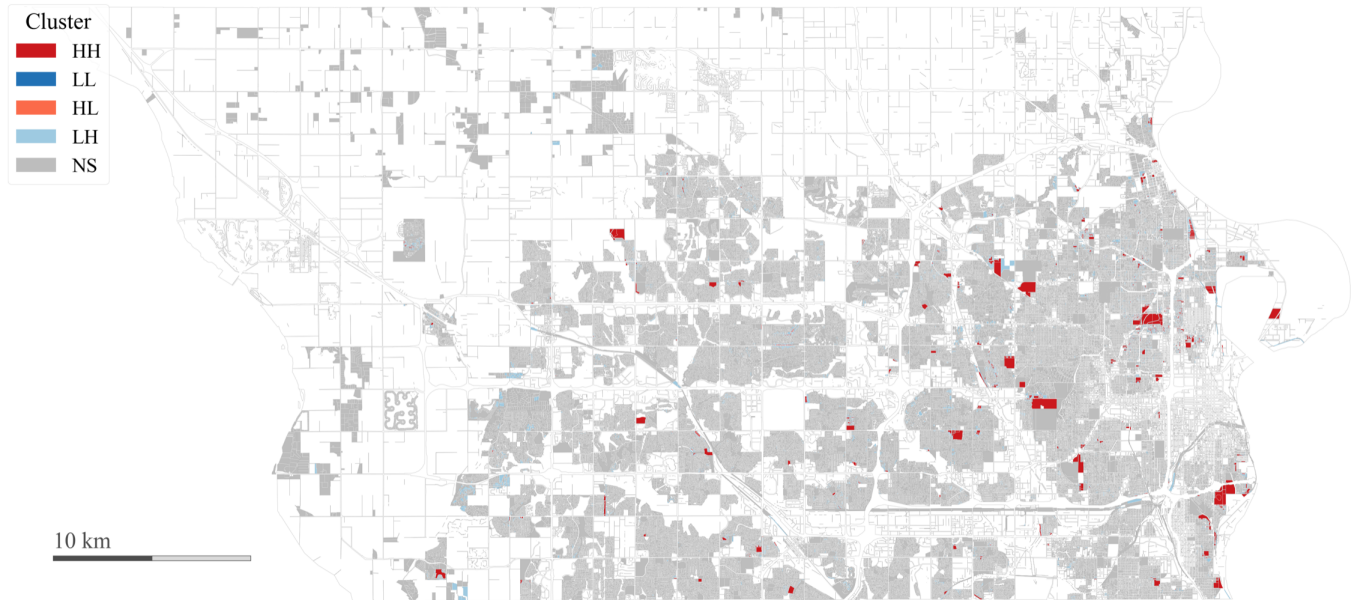


Figure 8: **Figure S8.** Government/Nonprofit Owners Spatial Clustering (LISA) by Parcel

Appendix Tables

Table S1: Alignment between clusters and production labels

Method	Clusters	Noise %	Weighted purity (overall)	Weighted purity (non-noise)	NMI	ARI
Kmeans (K=2)	2	0.00%	95.1%	95.1%	0.806	0.955
Kmeans (K=3)	3	0.00%	98.0%	98.0%	0.891	0.980
Kmeans (K=4)	4	0.00%	98.1%	98.1%	0.522	0.281
Kmeans (K=5)	5	0.00%	98.4%	98.4%	0.882	0.968
HDBSCAN (leaf)	44	46.16%	95.2%	99.0%	0.162	0.018

Table S2: Kmeans cluster composition by production label

Cluster	Size	Individual	LLC	Trust	Corporation	Gov/Nonprofit	Dominant label	Purity
0	143,362	143,358	4	0	0	0	Individual	100.0%
1	2,059	7	924	2	1,126	0	Corporation	54.7%
2	7,790	0	7,790	0	0	0	LLC	100.0%
3	5,482	688	3	4,790	0	1	Trust	87.4%
4	1,985	92	83	264	1,052	494	Corporation	53.0%

Table S3. Ownership classifier configuration and training

Component	Setting
Base model	BERT base uncased with 5-way Softmax head
Labels emitted	Individual; LLC; Corporation; Trust; Gov/Nonprofit
Training data	2,520 hand-labeled names (augmented to 5,040)
Loss & regularization	Focal loss; classification-head dropout = 0.20
Model selection	3-fold CV; early stopping on validation weighted F1
Reliability checks	Probability calibration; stratified human audit
High-confidence rule	p-hat max ≥ 0.80 used for robustness subset
Deployment scope	212,310 owner records countywide

Table S4. Production classification coverage and high-confidence shares (countywide)

Owner type	Count	Percent	High-confidence count	High-confidence coverage %
Individual	162,079	76.3	161,409	99.6
LLC	29,161	13.7	27,722	95.1
Corporation	5,588	2.6	4,169	74.6
Trust	7,006	3.3	6,844	97.7
Gov/Nonprofit	8,476	4.0	7,516	88.7
Total	212,310	100.0	207,660	97.8

Additional Tables (To Be Added)

Table S5. Robustness checks: Key contrasts under alternative specifications

Risk ratios (RR) and 95% confidence intervals for selected contrasts across model specifications. Reference group: Individual — Single in all models.

Group	Primary (M2)	With Distance (M4)	Attenuation
LLC — Single	1.670 [1.311–2.126]***	1.645 [1.293–2.093]***	1.5%
LLC — Multi	1.379 [1.104–1.722]**	1.358 [1.086–1.698]**	1.5%
Individual — Multi	1.209 [1.038–1.407]*	1.201 [1.031–1.399]*	0.7%
Trust — Multi	1.459 [1.043–2.041]*	1.446 [1.034–2.023]*	0.9%
Corporation — Multi	1.568 [0.960–2.561]	1.548 [0.947–2.531]	1.3%

Note: M2 = primary model (neighborhood FE, controls for log value and log acres); M4 = M2 + log(1+owner distance). Attenuation = percentage reduction in log(RR) from M2 to M4. Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Standard errors clustered by neighborhood. $N = 183,313$ (M2, M4).

Additional robustness checks: - High-confidence subset ($\hat{p} \max 0.80$): Key contrasts remain significant with directionally consistent estimates ($n = 179,000$) - 10% overlap SFHA rule: Results stable when relaxing majority-area threshold to 10% parcel overlap - Alternative SE specifications: Robust (HC3) and spatial HAC (Conley) standard errors yield consistent inference for primary contrasts

Table S6. Owner-level SFHA intensity model

This table addresses the concern that multi-property owners are mechanically more likely to own SFHA parcels simply by holding more parcels. The model estimates SFHA intensity at the owner level using a Poisson regression with an offset for log(total SFR parcels owned), effectively controlling for portfolio size.

Entity type	N owners	Mean SFHA parcels	IRR [95% CI]	p-value
Individual	139,154	0.010	1.000 (ref)	—
LLC	7,646	0.060	2.577 [2.318–2.866]	<0.001
Corporation	1,076	0.076	3.746 [2.997–4.682]	<0.001
Trust	5,628	0.016	1.644 [1.331–2.031]	<0.001
Gov/Nonprofit	1,098	0.060	2.867 [2.239–3.672]	<0.001

Note: Poisson GLM with log link; offset = log(total SFR parcels owned). $N = 154,602$ unique owners. IRR = incidence rate ratio. The offset specification means IRRs represent the rate of SFHA parcels per parcel owned, controlling for portfolio size.

Interpretation: After accounting for the mechanical effect of portfolio size, entity-owned portfolios still exhibit significantly elevated SFHA intensity relative to individual-owned portfolios. Corporations show the highest IRR (3.75 $\times$), followed by Gov/Nonprofit entities (2.87 $\times$), LLCs (2.58 $\times$), and Trusts (1.64 $\times$). These results complement the parcel-level analysis (Table 7) by confirming that the elevated SFHA exposure among entity owners is not simply an artifact of larger portfolio sizes.

Table S7. Eigenvector spatial filtering sensitivity analysis

This table presents results from a sensitivity analysis varying the number of eigenvector spatial filters (ESF) to assess whether ownership–SFHA associations are robust to explicit spatial modeling. Eigenvectors are extracted from a $k=8$ nearest-neighbor spatial weights matrix and included as controls in the Poisson GLM with neighborhood fixed effects (Griffith, 2003).

N Filters	Moran's I	LLC RR	LLC p-value	Corp RR	Trust RR	Gov/NP RR
0 (FE only)	0.392	1.379	<0.001	1.162	0.935	0.804
15	0.260	1.386	<0.001	1.096	0.936	0.782
30	0.312	1.285	<0.001	0.986	0.886	0.712

N Filters	Moran’s I	LLC RR	LLC p-value	Corp RR	Trust RR	Gov/NP RR
50	0.146	1.316	<0.001	0.991	0.879	0.622
75	0.146	1.316	<0.001	0.995	0.859	0.701
100	0.198	1.270	<0.001	0.946	0.878	0.619
150	0.126	1.225	<0.001	0.903	0.904	0.690
200	0.003	1.235	<0.001	1.012	0.916	0.665

Note: $N = 183,313$ SFR parcels. Reference group: Individual owners. Moran’s I computed on model residuals using an 8-nearest-neighbor weights matrix. RR = risk ratio for entity type (pooling single- and multi-parcel owners). With 200 spatial filters, residual Moran’s I decreases from 0.392 to 0.003—a 99% reduction in residual spatial autocorrelation. The LLC risk ratio attenuates modestly from 1.38 to 1.24 (approximately 10% attenuation) but remains statistically significant ($p < 0.001$) across all specifications, indicating that the LLC–SFHA association is robust to spatial structure.

Model Card — Ownership-Form Classifier

A1. Performance (evaluation set, $n = 756$)

Overall: Accuracy = 0.9934; Micro-F1 = 0.9934; Macro-F1 = 0.9800; Weighted-F1 = 0.9934.

Per-class (Precision / Recall / F1 / Support): - Corporation: 1.000 / 0.9286 / 0.9630 / 42 - Individual: 0.9982 / 0.9982 / 0.9982 / 561 - LLC: 0.9878 / 1.0000 / 0.9939 / 81 - Government/Nonprofit: 0.9429 / 1.0000 / 0.9706 / 33 - Trust: 0.9744 / 0.9744 / 0.9744 / 39

A6. Ablation & Baseline Comparisons

Token-removal ablation: From test strings, remove {LLC, LLP, LP, Inc, Corp, Ltd, Trust} (case-insensitive, punctuation-agnostic).

Rule-based baseline: Regex: CITY/COUNTY/STATE OF... → Government/Nonprofit; LLC/LLP/LP; TRUST/Trustee; INC/INCORPORATED/CORP/COMPANY/LTD; else Individual. Precedence: Government/Nonprofit → LLC → Trust → Corporation → Individual.

Split hygiene: Ablations evaluated on the held-out test set; augmented variants tied to the same base string remain in the same fold.

Overall performance comparison

Model	Accuracy	Macro F1	Micro F1	Weighted F1
BERT — Original	0.993386	0.979999	0.993386	0.993356
BERT — Ablated	0.953704	0.885115	0.953704	0.952532
Rule-based Baseline	0.841270	0.509683	0.841270	0.806666